Functional and Aesthetic Outcome of Extensive Orbital Floor and Medial Wall Fracture via Navigation and Endoscope-assisted Reconstruction

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Background and Purpose: Extensive orbital floor and medial wall fractures compared with isolated orbital wall fractures are more likely to require surgical correction because of a higher possibility of complications like diplopia, enophthalmos, or numbness. The unique and complex contours of the orbital anatomy limit the intraoperative view of the intraorbital anatomy, and complex orbital fractures involving the buttress of the transition zone area all make orbital reconstruction surgery more challenging. The aim of this study was to describe our experience with surgical approaches using navigation- and endoscope-assisted guidance for extensive orbital floor and medial wall fracture reconstruction.

Patients and Methods: A retrospective study was conducted on consecutive 17 patients from 2015 to 2017 presenting with unilateral extensive orbital floor and medial wall fractures at the Chang Gung Memorial Hospital, Linkou Branch. The fractures were treated surgically with a preformed mesh plate and layered Medpor (Porex Surgical Inc, Atlanta, Ga) through navigation and endoscopy. The preoperative and postoperative functional and aesthetic outcomes were described.

Results: All extensive orbital floor and medial wall fractures were successfully reconstructed. Of the 17 patients, 11 experienced diplopia preoperatively, and for 2 of the 11 patients, diplopia improved immediately after surgery. In the remaining 9 patients, diplopia still persisted after surgery; however, diplopia recovered after an average of 3.44 months (range, 1–9 months). Average enophthalmos among the 10 patients, evaluated by postoperative follow-up computed tomography scan, improved from 2.99 to 0.68 mm. There were no major complications during follow-up, and all patients were satisfied with their final appearance and function.

Conclusions: On the basis of the results, our surgical approach using preformed titanium mesh plates and Medpor under the assistance of navigation and endoscopy can be a safe, accurate, and effective method for the management of extensive orbital floor and medial wall fractures and clearly optimizes functional and aesthetic outcomes.

Key Words: extensive orbital blowout fracture, medial wall, orbital floor, preformed titanium plate, Medpor, navigation, endoscopy

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The 2 most common orbital fractures are orbital floor and medial wall fractures. Among orbital fracture characteristics, combined fractures of the floor and medial wall compared with isolated orbital wall fractures are more likely to require surgical correction because of increased volume expansion that may result in possible complications such as enophthalmos. In consideration of functional and aesthetic outcomes, orbital reconstruction is a demanding procedure. Orbital reconstruction surgery requires a thorough orbital defect assessment and precise restoration of orbital dimensions to prevent undesired clinical outcomes such as enophthalmos, diplopia, and cheek or nasal numbness.[1] For orbital reconstruction, it is important to consider the unique and complex contours of the orbital anatomy. Several studies have previously reported the reconstruction techniques in combined orbital medial and floor wall fractures involving the buttress of the transition zone.[2–5] The importance of reconstitution of the original buttress of the transition zone between the medial wall and the orbital floor should be considered.

Management of extensive orbital floor and medial wall fractures especially involving the transition buttress is arduous because of the loss of an anatomical reference point and the absence of bony structures to support implants. When the buttress of the transition zone was intact, it was used to support several individual implants over each of the fractures. However, when this support was compromised, the proper placement of implants became technically challenging for craniofacial surgeons because of the absence of a stable keystone.[5,6] Besides challenging anatomical reduction, limited intraoperative view of intraorbital anatomy via small conventional orbital access and dissection also increased the risk of optic nerve injury and uncontrolled bleeding.

With the advances of computer-aided design software and surgical navigation, planning the details of a surgical procedure in a 3-dimensional (3D) virtual environment and execution with real-time guidance can significantly improve precision and achieve satisfactory clinical outcomes. Since navigation-assisted orbital reconstruction was first introduced by Gellrich et al[7] in 2002, several authors have subsequently modified and refined these computer-aided techniques.[8–11] A recent systematic review demonstrated that surgical navigation is a very helpful tool when dealing with the reconstruction of such orbital fractures.[12]

Endoscopy has been applied to fractures of the zygomatic arch, orbit, frontal sinus, and condyle.[13,14] The benefits of endoscopy for orbital fractures include minimal access, maximum and safe visualization of the target and critical structures, in addition to equally accurate reconstruction and stable fixation compared with the conventional technique.[15] In particular, the endoscope-assisted method for approaching a medial orbital wall blowout fracture via a transcaruncular incision demonstrated to be safer for dissecting and identifying the herniated fat and posterior ledge to avoid accidently injuring the optic nerve using conventional blunt dissection.

In the present study, we describe our surgical strategy for the reconstruction of extensive orbital floor and medial wall fractures carried out at the Chang Gung Memorial Hospital from 2015 to 2017. Moreover, we report the improvements in functional and aesthetic outcomes via intraoperative navigation- and endoscope-assisted reconstruction, which allow more accuracy and safety during surgery.

MATERIALS AND METHODS

Study Design and Samples

This study was approved by institutional review board of the Chang Gung Memorial Hospital (201800605B0). A total of 17 patients were surgically treated for extensive orbital floor and medial wall fractures at the Chang Gung Memorial Hospital, Linkou Branch, from

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January 2015 to March 2017. Patient charts were reviewed according to the criteria designed for the study. Only patients who had undergone surgical correction in combination with navigation and endoscopy for traumatic orbital fractures performed by a single surgeon (H. T. Liao) were selected for the study. The outcomes for all patients and the computed tomography (CT)–measured enophthalmos of 10 of the 17 cases were described.

Inclusion Criteria

Inclusion criteria were as follows: (1) CT evidence of orbital fractures involving the orbital floor and the medial wall with buttress of the transition zone, (2) patients aged 18 and above at the time of surgery, (3) time interval between injury and surgery which was at least longer than 14 days, (4) preoperative and postoperative clinical and medical records, and (5) uniform technique. Patients who did not meet these criteria were excluded from the study.

Computer-Assisted Procedure

Preoperative spiral CT data sets for each patient with 1-mm collimation/slice thickness were used for preoperative planning and simulation. The BrainLAB 3D modeling software iPlan CMF (BrainLAB AG, Munich, Germany) was used for the Digital Imaging and Communications in Medicine data set processing and included object creation, mirroring of the unaffected side of orbital bone to the affected side, objective data measurement, and data export.[15]

The virtual planning data described above were loaded into the navigation computer preoperatively. The BrainLAB Vector Vision navigation platform was used for orbital reconstruction intraoperatively. In the preoperative setting up, a skull reference array (Latero Reference Star, BrainLAB) was fixed to the contralateral parietal bone with 1 self-drilling screw. This was followed by registration scanning of the frontal and periorbital skin area with a laser device (Z- touch, BrainLAB). Referencing accuracy was then verified by placing a navigation pointer at anatomic landmarks (upper central incisors). Deviation of less than 1 mm between the landmark and the preoperative CT confirmed successful registration (Fig. 1).

Surgical Technique

All patients were operated under general anesthesia. Preoperative antibiotic prophylaxis (cefazolin and gentamicin) were given perioperatively for 3 days.

Our surgical approach combined inferior transconjunctival and medial transcaruncular incisions to gain adequate access to the extensive orbital floor and medial wall fractures. If orbital fractures included involvement of zygoma fractures, we changed to subcilliary and medial transcaruncular incisions; if the orbital fracture was the only lesion involved in the inferior part of the medial wall, single incision with inferior transconjunctival or subcilliary incision was adequate for surgical access. The inferior oblique muscle was disinserted from the bone just posterior to the inferomedial rim with a periosteal elevator to correctly access the entire medial wall and orbital floor.

We used medial transcaruncular incisions for endoscopically assisted reconstruction of the concomitant medial wall and orbital floor fractures. We used a 2.7-mm, 30° telescope (Karl Storz, Tuttlingen, Germany) to visualize the medial orbital wall defect, and the herniated orbital content was retracted by the malleable retractor to create space for the endoscope. There were 2 main surgical timings indicating the use of the endoscope-assisted technique. First, we used the endoscope-assisted surgical approach in the dissection of the medical wall to prevent optic nerve injury and to have an adequate visualization of the surface landmark of the anterior and posterior ethmoid arteries. Second, the technique was used to verify whether the optic nerve was compressed by implants after we positioned the Medpor layers and the preformed titanium mesh plate (Fig. 2). After implant fixation, a normal forced duction test ensured that no entrapment of the extraocular muscle and periorbita had occurred.

We used preformed titanium mesh plates in all cases with extensive orbital floor and medial wall fractures. Additional mesh plate may be required when the lateral portion of the orbital floor cannot be covered by preformed titanium mesh. In most cases, the preformed anatomic orbital plates could not guarantee anatomic reconstruction as the collapse of the transition zone at the junction of the floor and medial wall frequently occurs in large 2-wall fractures. Thus, we layered additional multiple pieces of 1.5-mm-thick Medpor (Porex Surgical Inc, Atlanta, Ga) under or medially to the preformed mesh plate to reinforce support of the preformed mesh plate especially at the orbital buttress of the transition zone region until the position of the mesh plate was stable enough on the planned position confirmed by navigation. Throughout the surgery, repeated navigation pointer check ensured that the orbital reconstruction was proceeding precisely and safely, matching the boundaries of the mirrored presurgical plan representing the ideal outcome (Fig. 3).

Postoperatively, intravenous dexamethasone 5 mg every 6 hours was routinely administered for 1 day if there were no contraindications, and ice packing of the injured orbit was performed for 1 week to reduce globe edema. Postoperative follow-up CT scans were obtained for patients at least 6 months after surgery. We evaluated functional and aesthetic outcomes of patients at the 1-week, 3-week, 6-week, 3-month, 6-month, 9-month, and 12-month follow-ups (Figs. 4–6).

Study Variables and Outcome Parameters

Study variables included the following: patient age, sex, cause of injury (eg, motor vehicle accident, assault, sport, fall), associated fracture characteristics, and history of previous orbital reconstruction

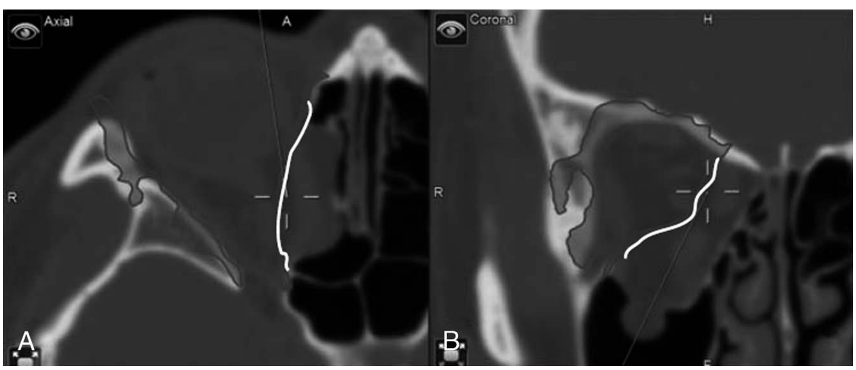

FIGURE 1. Navigation-assisted reconstruction. The axial (A) and coronal view (B) of the CT image showed the tip of the navigation pointer placed at the supposed repair position (in relation to the white-colored mirror image of the unaffected orbit) of the right extensive orbital floor and medial wall fracture. A large defect extending to the posterior part of medial wall was demonstrated in the axial view (A).

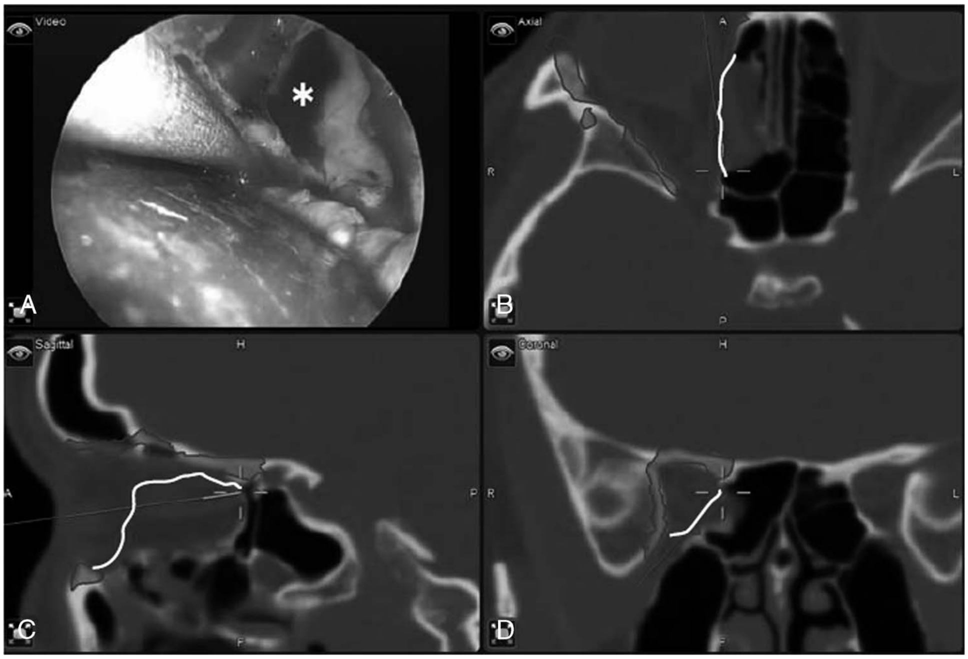

FIGURE 2. Endoscopic-assisted orbital medial wall dissection. A right medial wall orbital defect (white asterisk) is shown in an endoscopic image. The tip of the navigation pointer was placed at the distal limit of the dissected field of the medial wall (A), and its position in relation to the mirror image was demonstrated in the axial (B), sagittal (C), and coronal views (D) of the CT image.

surgery. Outcome parameters included the following: preoperative and postoperative diplopia in all patients and preoperative and postoperative CT analyses of enophthalmos in 10 of 17 patients.

Measurement of Diplopia

Diplopia was considered clinically present when double vision was reported in any direction of gaze, even in extreme gaze, regardless

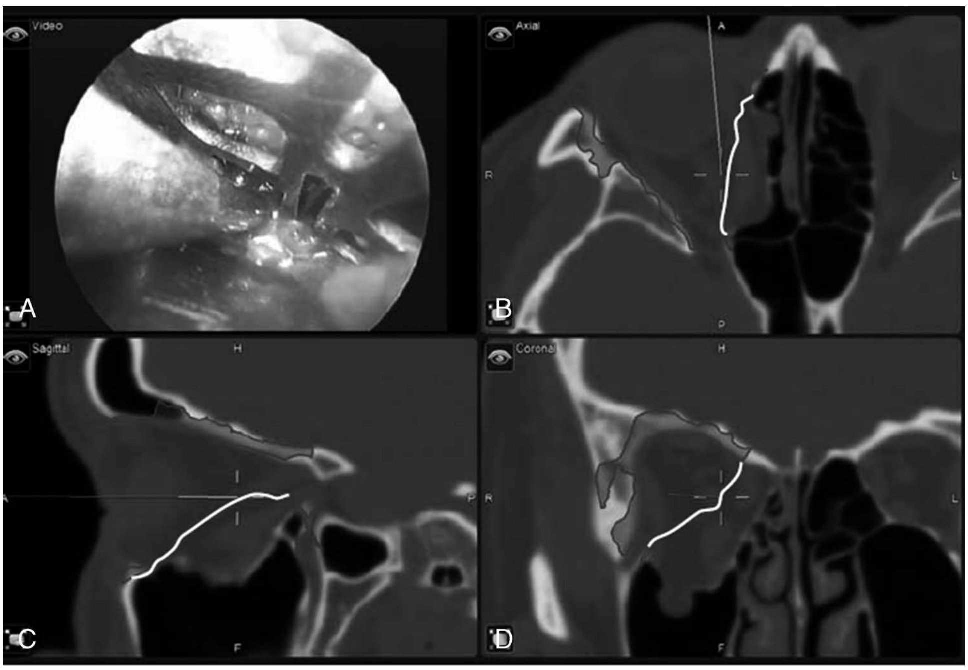

FIGURE 3. Reconstructed right orbital medial wall fracture. A right orbital medial wall fracture reconstructed with a preformed titanium mesh plate supported by the pieces of Medpor medially is shown in the endoscopic image (A). The tip of the navigation pointer was placed at the corresponding position of the stable point in the axial (B), sagittal (C), and coronal CT views (D). The images show an ideal outcome, with the boundaries of the reconstructed orbital floor and medial wall matching the ones in the white-colored mirrored presurgical plan.

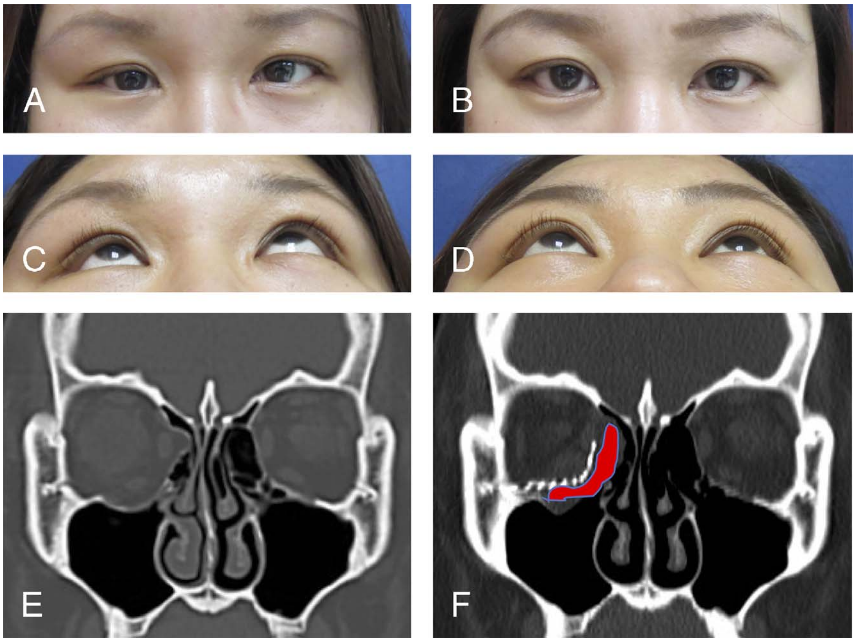

FIGURE 4. Facial CT scans and photographs of case 1. This 29-year-old female patient was beaten by others, presenting with severe enophthalmos and hypoglobus. She was diagnosed with right orbital floor and medial wall fracture with comminuted buttress of the transition zone. She received orbital reconstruction with a preformed mesh plate supported by layered Medpor through navigation and endoscopy. The results demonstrated a profound improvement of right side enophthalmos. A, Preoperative front view; B, postoperative front view; C, preoperative inferior view; D, postoperative inferior view; E, preoperative coronal view of CT scans; F, postoperative coronal view of CT scans (grey color block demonstrated where we placed the Medpor).

of severity. Completely resolved diplopia was defined as the absence of diplopia in all visual fields.

Measurement of the Enophthalmos

To minimize measurement errors, slice orientation was standardized using the 3D multiplanar reconstruction function to the reference Frankfurt horizontal plane.[16] We measured the location of the eyeball on transverse CT scans that were taken on frontal gaze, clearly visualizing the center of both eyeballs and maximizing the size of both eyeballs containing the corneal apex. The line extending between the lateral orbital rim and the medial orbital rim was defined as line A. The line extending from the corneal apex to line A via the center of the eyeball was

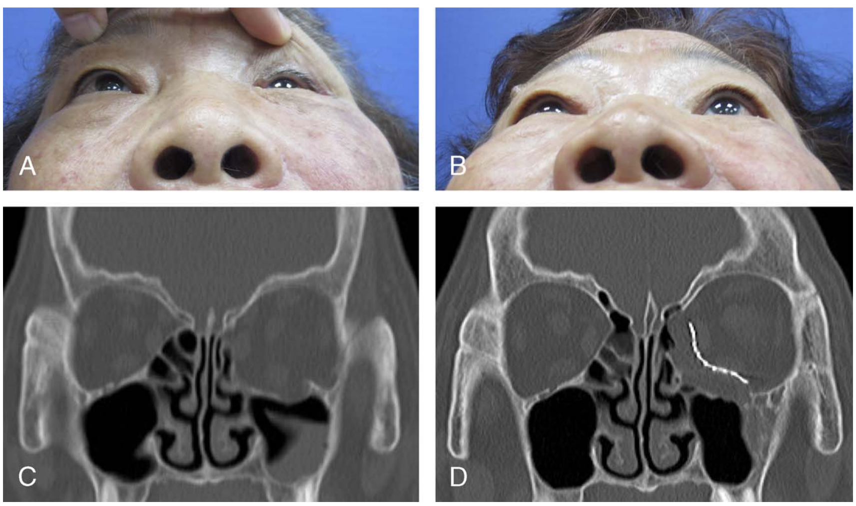

FIGURE 5. Facial CT scans and photographs of case 2. This 72-year-old female patient presented with severe enophthalmos and hypoglobus after a traffic accident. She was diagnosed with left orbital floor and medial wall fracture with comminuted buttress of the transition zone. She received orbital reconstruction with a preformed mesh plate and layered Medpor through navigation and endoscopy. The postoperative results demonstrated a profound improvement of left side enophthalmos after corrective surgery. A, Preoperative inferior view; B, postoperative inferior view; C, preoperative coronal view of CT scans; D, postoperative coronal view of CT scans.

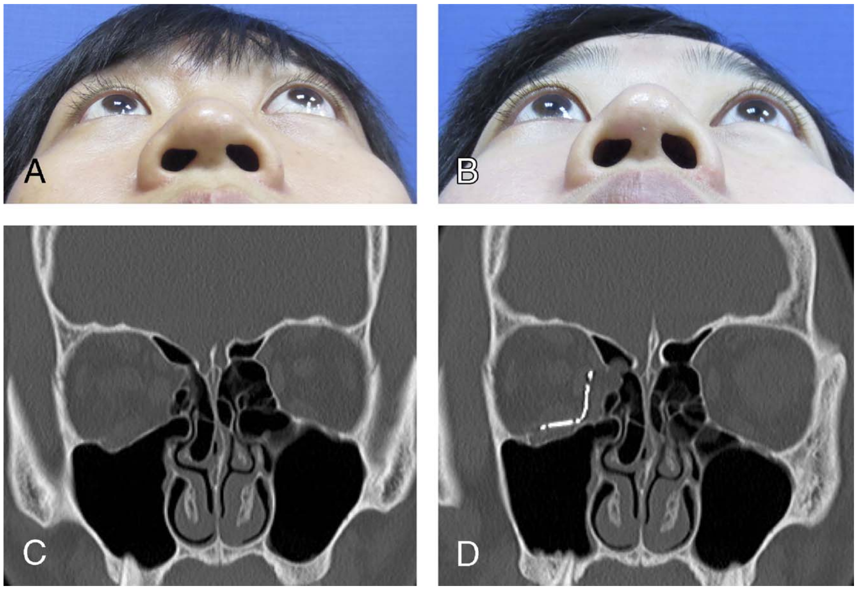

FIGURE 6. Facial CT scans and photographs of case 3. This 20-year-old female patient was hit on her right eye by her friend, presenting with enophthalmos and hypoglobus. She was diagnosed with right orbital floor and medial wall fracture with comminuted buttress of the transition zone. She received orbital reconstruction with a preformed mesh plate and layered Medpor through navigation and endoscopy. The preoperative results demonstrated a profound improvement of right side enophthalmos after corrective surgery. A, Preoperative inferior view; B, postoperative inferior view; C, preoperative coronal view of CT scans; D, postoperative coronal view of CT scans.

defined as line B. The globe position was measured from the difference in the value of B. We defined enophthalmos as the difference in the distance between the unaffected eye and the affected one[17] (Fig. 7).

Statistical Analysis

The results data were processed and analyzed as descriptive statistics including mean values, ranges, and frequencies. The paired *t* test was used to calculate the differences between the preoperative and postoperative enophthalmos ($P < 0.001$ was considered statistically significant).

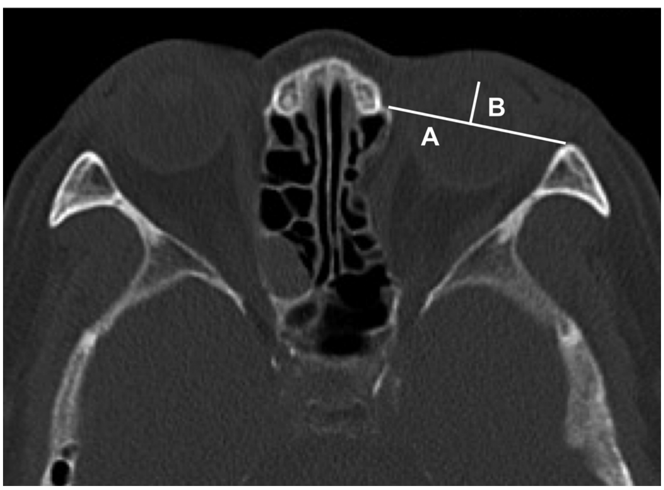

FIGURE 7. Measurement of enophthalmos. To measure the degree of enophthalmos in the preoperative and postoperative axial views of the CT image, we defined line A, the distance between the lateral orbital rim and the medial orbital rim, and line B, the line extending from the corneal apex to line A via the center of the eyeball. Enophthalmos was defined as the difference in line B between the unaffected orbit and the affected one.

RESULTS

A total of 17 patients were enrolled in the study. All patients were treated with orbital reconstruction surgery via intraoperative navigation and endoscope technique. Computed tomography scans were performed in all patients preoperatively. Postoperative follow-up CT scans were performed in 10 of the 17 patients. All patients were identified for data collection and outcome analysis. Ages ranged between 18 and 85 years. Patient demographics, cause of injury, associated fracture characteristics, history of previous orbital reconstruction surgery, time from injury to surgery, and follow-up are summarized in Table 1.

With regard to the surgical approach, medial transcaruncular and inferior transconjunctival incisions were used in 9 (52.9%) cases, a single subcilliary incision was used in 3 (17.6%) cases, subcilliary and medial transcaruncular incisions were used in 3 (17.6%) cases, and a single inferior transconjunctival incision was used in 2 (11.8%) cases. A preformed titanium mesh plate and Medpor were used in 8 (47.0%) cases; a preformed titanium mesh plate, Medpor, and titanium mesh plates were used in 5 (29.4%) cases; a preformed titanium mesh plate was used in 2 (11.8%) cases; and a preformed titanium mesh plate and titanium mesh plates were used in 2 (11.8%) cases. There were no major postoperative and intraoperative complications recorded. Only 3 patients had minor postoperative complications. One patient had transient ectropion and hematoma, 1 had transient ectropion, and 1 had transient entropion. No revision surgery was required in the patient group. Table 2 summarizes the surgical demographics for all patients.

Table 3 summarizes the diplopia outcome. Of the 17 patients, 11 had diplopia preoperatively. In 2 of the 11 patients, diplopia improved immediately after surgery. In 9 of the 11 patients, diplopia persisted after surgery, and diplopia recovered after an average of 3.44 months (range, 1–9 months). Of the 17 patients, 6 did not exhibit diplopia preoperatively nor postoperatively.

Assessment of preoperative and postoperative enophthalmos is summarized in Table 4. All cases of preoperative enophthalmos

TABLE 1. Summary of Demographics of Patients (N = 17)

Characteristics	No. Patients
Age, y	33.65 ± 20.37* (18–85)†
Sex, n (%)	
Male	9 (52.9)
Female	8 (47.1)
Smoking	
Current	7 (41.2)
Never	10 (58.8)
Hypertension	
Yes	2 (11.8)
No	15 (88.2)
Diabetes mellitus	
Yes	2 (11.8)
No	15 (88.2)
Cause of injury	
MVA	13 (76.5)
Assault	2 (11.8)
Fall	2 (11.8)
Sport	0 (0.0)
Location	
Left	13 (76.5)
Right	4 (23.5)
Associated fractures	
Zygomatic fracture	6 (35.3)
Lefort I fracture	5 (29.4)
Lefort II fracture	2 (10.0)
NOE type 1 fracture	2 (11.8)
NOE type 2 fracture	0 (0.0)
Mandible fracture (angle)	1 (5.9)
Mandible fracture (body)	0 (0.0)
Mandible fracture (condylar)	0 (0.0)
Frontal bone fracture	1 (5.9)
Nasal fracture	1 (5.9)
History of previous orbital reconstruction surgery	
Yes	4 (23.5)
No	13 (76.5)
Time from injury to surgery, d	122.94 ± 112.27 (14–366)
Follow-up, mo	7.89 ± 5.78* (1.22–24.94)†

*Mean ± SD. †Range.

MVA, motor vehicle accident; NOE, nasoorbitoethmoidal.

achieved visibly symmetric eyeball projection after surgery. Only 10 of the 20 patients underwent CT for postoperative analysis. The average enophthalmos of the 10 patients improved from 2.99 mm (range, 2.03–3.72 mm) to 0.68 mm (range, 0.12–1.29 mm). The paired *t* test revealed a highly significant ($P < 0.001$) improvement in average enophthalmos across all patients.

DISCUSSION

We considered that the complexity of the anatomical region and the limited intraoperative view may lead to unsatisfactory surgical outcomes and even failure of reconstruction. The vital areas of the orbital anatomy include the buttress of the transition zone, the inferomedial bony ledges, the S shape of the orbital floor in the posterior third region, and the inferior two thirds of the medial wall presenting a bulge toward the orbit.[18,19]

The challenge of repairing extensive orbital floor and medial wall fractures lies particularly in the most posterior part of the buttress of the transition zone, between the orbital floor and the medial wall. However, the buttress of the transition zone is difficult to visualize intraoperatively. In consideration of potential risk of optic nerve injury, it becomes even more difficult to visualize in dissection further posteriorly. Reconstruction of the transition buttress between the medial orbital wall and the floor is essential and allows to determine the further positioning of the orbital contents and the globe. Unstable reconstruction of the buttress of the transition buttress may lead to an undesirable secondary change of the orbital contour and, as a consequence, may lead to the development of subsequent residual enophthalmos.

Several techniques have been described to deal with combined orbital floor and medial wall fractures.[4–6,20–28] Several researchers have reconstructed the orbital floor and medial wall concomitantly using multiple small pieces of implants. Choi et al used[20] 2 separate porous polyethylene channel implants (thickness, 2.3 mm), fixed to the orbital rim by titanium plates placed within the longitudinal channels. In their study, the implants used were relatively thick and may have potentially caused postoperative hyperglobus. Su and Harris[6] repaired combined orbital walls by overlapping 2 or 3 nylon foil implants through combined transcaruncular-transconjunctival incisions. In their study, the buttress of the transition zone was not involved in most of the patients, and the suspected instability of the buttress of the transition zone area was likely caused by the use of a nonrigid implant. Hur et al[5] described the use of multiple pieces of rigid implants instead of 1 large implant, and sequential repair first of the floor and then of the medial wall, with a focus on the reconstruction of key areas. Their method of inlay implantation for medial wall repair, filling the ethmoidal sinus with several pieces of porous polyethylene, was similar to our surgical strategy. Some surgeons have repaired the combined orbital wall fractures using a single large implant. Nunery et al[23] used a single large nylon foil implant (thickness, 0.4 mm) via a transcaruncular-transconjunctival approach. In their study, the wide exposure for the placement of the single large implant might have led to severe edema or hematoma postoperatively.

With advances in computer-aided design and manufacturing, several authors have described the successful repair of extensive orbital floor and medial wall fractures with personalized preformed implants.

TABLE 2. Summary of Surgical Demographics of Patients (N = 17)

Characteristics	No. Patients
Surgical approach, n (%)	
Medial transcaruncular + inferior transconjunctival	9 (52.9)
Subcilliary	3 (17.6)
Subcilliary + medial transcaruncular	3 (17.6)
Inferior transconjunctival	2 (11.8)
Type of implant	
Preformed titanium mesh plate + Medpor	8 (47.0)
Preformed titanium mesh plate + Medpor + titanium mesh plate	5 (29.4)
Preformed titanium mesh plate + titanium mesh plate	2 (11.8)
Preformed titanium mesh plate	2 (11.8)
Postoperative complication	
Transient ectropion	2 (11.8)
Transient entropion	1 (5.9)
Hematoma	1 (5.9)
Navigation	17 (100.0)
Endoscopy	17 (100.0)
Surgical revision	0 (0.0)

TABLE 3. Preoperative and Postoperative Functional Outcome: Diplopia

Preoperative Presentation, n (%)		1 Week After Surgery, n (%)		Time for Recovery, Mo
Diplopia (+)*	11 (64.7)	Diplopia (+)	9 (81.8)	3.44 ± 2.74† (1–9)‡
		Diplopia (−)	2 (18.2)	-
Diplopia (−)§	6 (35.3)	Diplopia (+)	0 (0.0)	-
		Diplopia (−)	6 (100.0)	-

*Presence of diplopia. †Mean ± SD. ‡Range. §Absence of diplopia.

Schon et al[22] reported that these individually designed preformed implants proved to be less time-consuming, more precise, and less invasive, compared with freehand bent implants for the reconstruction of orbital fractures. Kim et al[28] described a patient-specific puzzle implant for combined orbital floor and medial wall fractures involving the inferomedial strut (buttress of the transition zone). However, both studies were limited by the absence of cost-effectiveness analysis. Although patient-specific implants indeed provided more accurate restoration of the internal orbit and minimize the need for major intraoperative manipulations,[29] their fabrication was expensive and time-consuming.[30] Vehmeijer et al[31] reported that a 3D printed titanium orbital implant could cost between €3000 and €5000.

Compared with previously reported studies, our study revealed important benefits. We assumed that extensive damage to key orbital landmarks could severely affect the accuracy of conventional reconstruction.[32,33] Therefore, we used additional multiple pieces of Medpor to reinforce the support of the preformed mesh plates at the orbital buttress of transition zone areas. We used intraoperative navigation to provide real-time intraoperative objective guidance for the optimal positioning of the implants to achieve an accurate reconstruction of the orbital defect and to grossly minimize surgical complications. We also applied an intraoperative endoscopic technique to achieve improvement of adequate visualization and surgical safety.

The anatomical significance of the buttress of the transition zone has been discussed previously. Orbital reconstruction is particularly difficult when the buttress of the transition zone is missing due to the loss of an anatomical reference point for the medial wall and orbital floor reconstruction. Thus, we used a preformed mesh plate to reconstruct the buttress of the transition zone areas. If a collapse of the transition zone area occurred near the optic canal, which is less than 5 to 6 mm from the posterior ethmoid artery, and to avoid possible injury to the optic nerve, implants were not placed deeper toward the most posterior part.[34] On the basis of our experience, the implants were unstable if there was no adequate support for the distal part of the preformed mesh plate. To reinforce the support of the preformed mesh plate, we layered additional multiple pieces of 1.5-mm-thick Medpor. Otherwise, if only the anterior or the midportion of the transition zone area showed signs of collapse, it was not necessary to place additional layers of Medpor. We detected the stability and possible displacement of implants using intraoperative navigation. If it were considered that the reconstruction of transition zone would be unstable, we would place additional Medpor to provide further support.

TABLE 4. Preoperative and Postoperative Aesthetic Outcomes Among 10 Patients: Enophthalmos

No. (n = 10)	Preoperative, mm	Postoperative, mm	*t*	*df*	*P*
Enophthalmos	2.99 ± 0.53* (2.03–3.72)†	0.68 ± 0.40 (0.12–1.29)	11.284	9	<0.001

t, *t* score; *df*, degrees of freedom.
*Mean ± SD. †Range.

There have been several studies indicating that transition area defects between the floor and the medial wall especially benefit from navigation-assisted precision procedures.[21,33,35] Using navigation, the mirrored normal globe provided a valuable intraoperative navigation gauge to ensure that the orbital volume had been adequately restored to correct any enophthalmos and avoid postoperative residual enophthalmos.

Intraoperative navigation provided improvement of precision and surgical safety even for complex orbital reconstruction. A recent systematic review has indicated that surgical navigation was a very helpful tool when dealing with complicated orbital fractures.[12] In our patient group with such orbital fractures, these exhibited more variability in the defect shape, and failure to place the implant in the correct position might lead to atrophy, contractions, herniated tissue, and secondary undesirable clinical outcomes. In our patients, the orbital walls were often comminuted, and the bone fragments were often displaced into the maxillary sinus. In this context, the reconstruction of the absent pieces of bone was essential to support the globe and to restore orbital shape. However, a conventional assessment of orbital reconstruction based on experience and mainly subjective naked eye evaluation of symmetry and dimensions makes outcomes inherently unpredictable.[33,36] Hammer and Prein[32] reported that conventional techniques have resulted in increased orbital volumes in approximately 8.5% of traumatic orbital cases. In addition, accuracy became more unattainable in delayed or secondary orbital reconstruction attempts compared with primary reconstruction because of malunion, nonunion, bony absorption, loss of soft tissue envelope, and scarring.[33,37,38]

Navigation took guesswork and individual bias out of orbital reconstruction particularly in complex orbital fracture reconstruction. It provided real-time intraoperative objective guidance to achieve symmetrical and accurate reconstitution of the orbital defect. Several large series studies found successful improvement of enophthalmos, diplopia, and facial symmetry through navigation, particularly in delayed and secondary cases.[7,9,39–43] More importantly, safety was also enhanced because anatomical location and proximity to critical structures and landmarks can instantly be determined from navigation images.[44]

Orbital working spaces were crowded with vital structures, making it difficult to access and visualize the darker deeper orbit. Besides navigation, we used endoscope-assisted reconstruction to achieve adequate visualization and further improvement of surgical safety.

The well-known 24-12-6 generalization describing the distances between the anterior lacrimal crest, anterior ethmoidal foramen, posterior ethmoidal foramen, and optic canal was regarded as a useful localization tool.[45] Chen et al[34] reported that great care must be taken to avoid an iatrogenic injury to the optic nerve after coagulation of posterior ethmoidal artery because the optic foramen was located only 5 to 6 mm. Through adequate visualization via the endoscope technique, we could clearly detect the posterior ethmoidal artery and stop dissection in order not to endanger the optic nerve at the optic foramen. Any instrumentation beyond the posterior ethmoidal foramen could result in mechanical injury to the optic nerve.

A recent systemic review explicitly concluded that the clinical choice for the implant material characteristics was based on (1) stability and fixation, (2) contouring abilities, (3) biological behavior, (4) drainage, (5) donor site morbidity, (6) radiopacity, and (7) availability and cost-effectiveness.[46] Furthermore, materials for reconstructing the orbit were selected based on the requirement that the defect matched the mechanical properties of the material.[47]

In our patient group with extensive orbital floor and medial wall fractures, the accurate reconstruction of larger defects was our primary consideration. For the reconstruction of larger defects, a more rigid material was required. Compared with bone graft, and porous polyethylene sheets, titanium meshes had both better stability and fixation characteristics. Dubois et al[48] assessed that titanium could be an ideal implant for covering large anatomical defects and globe malposition if an implant-stabilizing surrounding bone or a distal landmark such as bone ledge was absent. In addition, several studies concluded that fixation was required to prevent migrations, which may lead to infections, fibrosis, and scarring, and may incidentally result in diplopia and even blindness. If the orbital rim was comminuted, fixation may be more complex. A titanium mesh may help secure the bony pieces.[49,50] Moreover, titanium meshes had radiopacity characteristics, which was useful for postoperative confirmation of the position of implant placement and postoperative analysis. In conclusion, titanium was biocompatible, had excellent physicomechanical properties and radiopacity, and was strong, rigidly fixable, widely available, relatively familiar to surgeons, and could be osseointegrated with minimal foreign body reaction.[51]

In large defects, the implant contour also becomes an increasingly important factor for repositioning the globe into the correct position.[52] In recent studies, patient-specific implants offered the best reconstructive contour, but these were expensive and time-consuming to fabricate. Freehand bent implants had the advantage of being cost-effective with years of clinical application, but their contour accuracy and ease of use were comparatively inferior. Strong et al[30] demonstrated that an evaluation of the orbital floor contour resulted in the greatest accuracy for patient-specific molding, followed closely by preformed mesh and finally by freehand bending. However, there was no statistical difference in orbital volume repair with any of these 3 techniques studies. A prospective multicenter study indicated that orbital reconstruction was significantly more precise when individualized implants were used compared with standard preformed implants. However, the authors also reported that an overlap in the use of individualized implants and navigation made it difficult to attribute the improved precision to a single factor.[53] On the basis of previous studies, we believe that preformed mesh implants are more advantageous with very good contour accuracy, moderate cost, and excellent ease of use for the current clinical application.

CONCLUSIONS

Treatment of orbital fractures, especially complicated ones, can often be very challenging. Our surgical approaches with preformed titanium mesh plates and Medpor under the assistance of navigation and endoscopy can be a safe, accurate, and effective method for the management of extensive orbital floor and medial wall fractures and clearly optimize functional and aesthetic outcomes.

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